

Department Of Computer Science

AI-02

OOPs Project Proposal

Hotel Management System in C++



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Introduction

The Hotel Management System is a C++ software application designed to automate hotel operations such as room booking, customer management, billing, and check-in/check-out.

It reduces :

- Human errors
- Slow processing
- Data mismanagement
- Difficulty in handling customer records

This project aims to solve these problems using Object-Oriented Programming (OOP) concepts in C++.

Objectives

The objectives of this project are:

1. To develop a hotel management system using C++.
2. To implement Object-Oriented Programming concepts.
3. To automate room booking and billing processes.
4. To maintain customer and room records efficiently.
5. To reduce manual work and improve data accuracy.

Problem Statement

Manual hotel management systems are inefficient because they:

- Consume more time

- Increase chances of errors
- Make record management difficult
- Reduce operational efficiency

Therefore, a computerized system is required to:

- Manage hotel rooms
- Store customer information
- Automate booking and billing
- Improve hotel management efficiency

OOP Concepts Used

OOP Concept	Usage in Project
Class	Room, Customer, Booking
Object	Customer and room objects
Encapsulation	Secure customer information
Inheritance	Different room categories
Polymorphism	Payment methods
Abstraction	Simplified system operations

Proposed Modules

1. Room Management Module
2. Customer Management Module
3. Booking Module
4. Billing Module

5. Admin Module

Functional

The system should:

- ☒ Add hotel rooms
- ☒ Display available rooms
- ☒ Store customer records
- ☒ Book rooms
- ☒ Generate bills
- ☒ Checkout customers
- ☒ Search room/customer data